

Divisible Workspace Blueprint

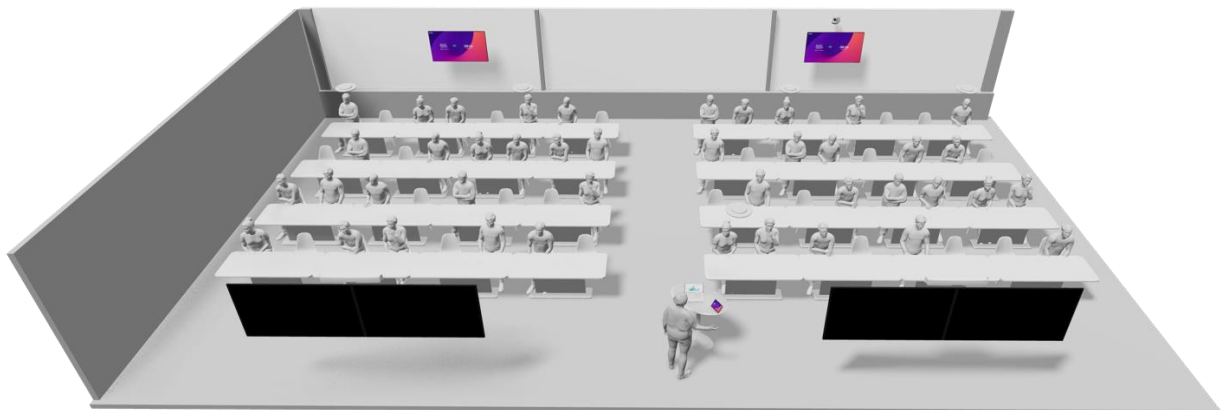
Network Restricted Installation Guide

This document provides the installation instructions for the Divisible Workspace Blueprint. This blueprint leverages Cisco Pro Series Microphones, Catalyst Switches and Cisco Video Devices to provide a simplified, easy to deploy, scalable offering for Divisible Workspaces.

This guide is applicable to any Premise Registered device that is deployed in a **Network Restricted fashion.**

Cloud registered devices should follow this guide: [Cloud Installation Guide](#)

Premise registered devices should follow this guide: [Premise Installation Guide](#)



All other documentation is available on GitHub:

<https://cs.co/divisibleworkspaceblueprint>

Best effort and community level support provided via a Webex messaging space.

You can join using this URL: https://eurl.io/#nakTe_Vn3

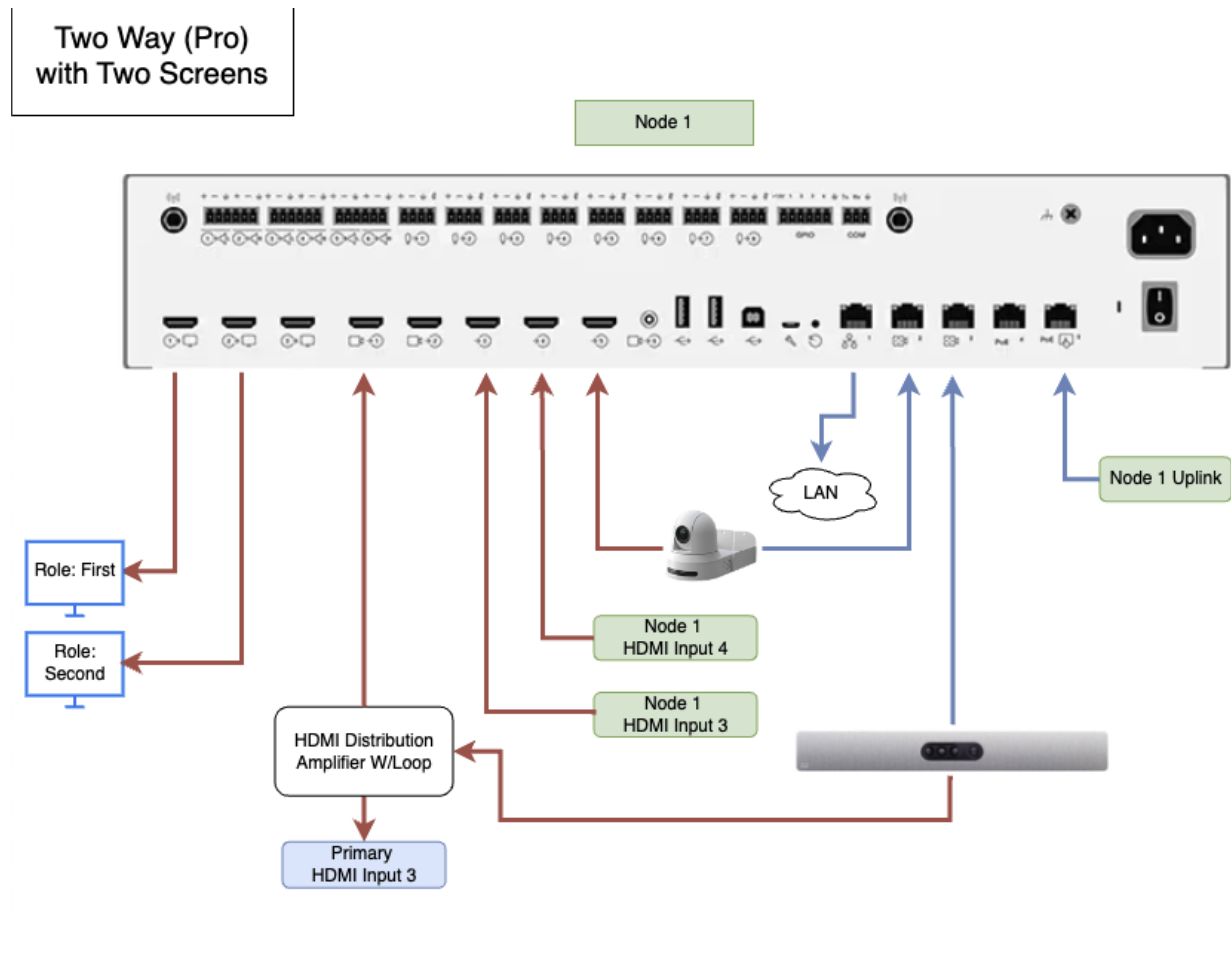
You can find the User Guide documentation here: [User Guide](#)

In-Room Installation

Physical Wiring

The blueprint provides several validated wiring schematics based on the Codecs selected, number of secondary room displays and the Catalyst 9K Switch.

Example Diagram:



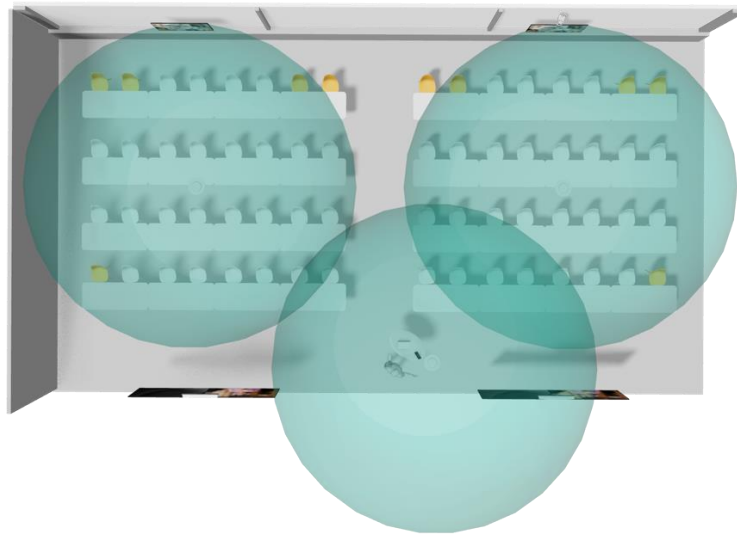
IMPORTANT

*Do **not** power on any Codecs, switch or connect Scheduler Panels until expressly instructed. The Blueprint leverages several mechanisms that rely on a pre-defined order of operations.*

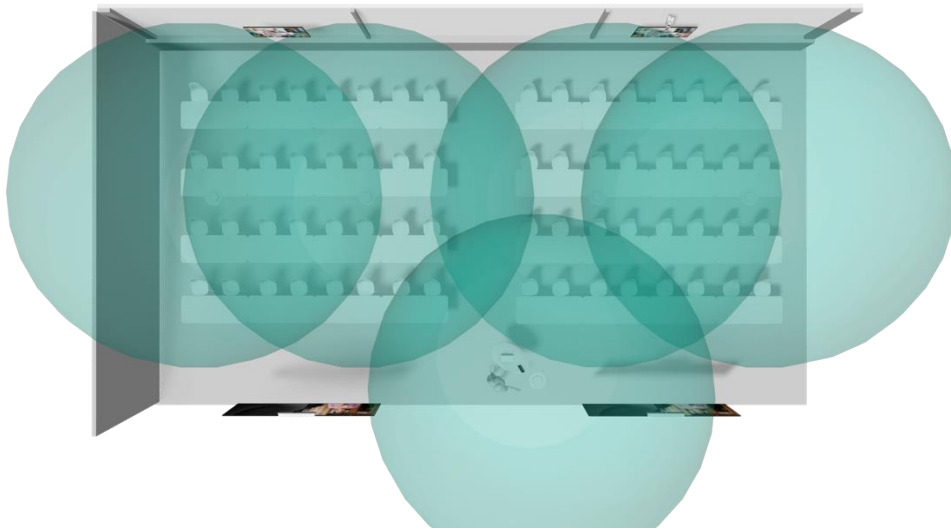
Microphone Installation Guidance

When installing Ceiling Microphone Pros you will want to follow these core concepts:

1. Center the Audience Microphones over the middle of the audience area.



2. If required for room width, add a second microphone to the audience area.



3. If you are leveraging a Ceiling Microphone Pro as the presenter Microphone, it should be positioned towards the middle of the combined space as close to the center line area as possible. This allows for adequate coverage of the “Stage” area configured for Presenter Tracking.



Presenter Track Example Perspective

Configuration Tasks

*Please do **not** power on your Codecs until **Step 3**.
If you are deploying Schedulers, do **not** connect them until **Step 4**.*

Step 1: Prepare the USB Key

Format a USB Key with a **4GB FAT32** partition.

Download the appropriate “**ciscortr.cfg**” file to match the n-way and switch model you are deploying from the GitHub Repository: [Switch Configurations](#)

Place the “**ciscortr.cfg**” file into the **root directory** of the USB Key.

*Please do **not** rename the file or place it into any subfolder.*

Step 2: Configure the Switch

Insert the USB key into the front port of the **Catalyst 9K Series**.

Power on the switch. Do **not power on the Codecs.**

*Switch auto-configuration will take **6 minutes**.*

IMPORTANT: The switch configuration is **NOT** automatically written to memory. This is handled by the macro install process later in this guide.

Please do not power off the switch until setup is complete.

Step 3: Register Codecs

Power on your codecs.

Register them using the initial start-up wizard selecting either cloud or premise deployment to match your installation.

Step 4: Connect Scheduler Panels (Optional)

If deploying Scheduler Panels, connect the additional Room Navigators to their appropriate ports on the Catalyst 9K switch.

Step 5: Configure Local Admin Users

Log into your **Node** Codec using its IP Address in your web browser. The default username is **admin** with a **blank** password. If you are asked to set a password, please do so now.

On the left-hand Menu, Select **Users** then click **Create User**.

Enter a **Username**, Select the Role as Admin, Enter the passphrase, confirmation and the password you set for the **admin** user above. Ensure the “Require passphrase change on next user sign in” is **Unchecked**.

Add New User

The screenshot shows the 'Add New User' form with the following fields and options:

- Username:** A text input field with a blue arrow pointing to it.
- Roles:** A group of radio buttons with 'Admin' selected. Other options are Audit, RoomControl, Integrator, and User. A blue arrow points to the 'Admin' radio button.
- Status:** A group of radio buttons with 'Active' selected. Other option is Inactive. A blue arrow points to the 'Active' radio button.
- Client Certificate DN:** A text input field.
- Client Certificate DN Help:** A text block explaining the format for client certificates: "If using client certificates for authentication, enter the client certificate's full Distinguished Name. Both the /CN=alice/DC=example/DC=com and the CN=alice, DC=example, DC=com formats are supported."
- Require passphrase change on next user sign in:** An unchecked checkbox.
- New passphrase:** A text input field with a blue arrow pointing to it.
- Generate new passphrase...:** A link with an information icon.
- Confirm passphrase:** A text input field.
- Your passphrase:** A text input field with a blue arrow pointing to it.
- When creating or modifying admin users, you must enter your own passphrase for verification.** A note.
- Create User:** A button.

Click **Create User**.

(Required) Repeat Step 5 on ALL Node Codecs
Can also be **optionally** applied on the Primary Codec

Step 6: Set the baseline Codec Configurations

Log into your **Node** Codec using its IP Address in your web browser.

On the left-hand menu, select **Settings**.

Modify the following configurations to the provided values:

Attribute	Value
HttpClient > AllowInsecureHTTPS	True
HttpClient > Mode	On
Macros > Mode	On
Video > Input > CameraConfigMode	Auto

(Required) Repeat Step 6 on ALL Node Codecs and the Primary Codec

Step 7: Download the Macro files

Download all the Files from the Macro directory of the “main” branch.

You should have the following files:

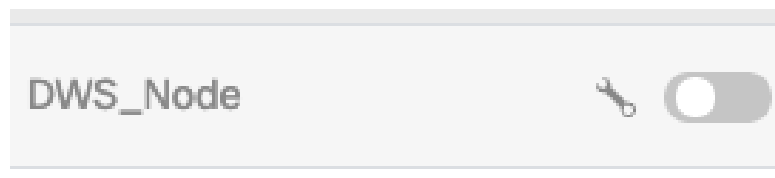
DWS_Setup.js, DWS_Audio.js, DWS_Core.js, DWS_Images.js and DWS_Node.js.

Step 8: Deploy Macro on the Node Codec(s)

Log in your **Node** Codec(s) using its IP Address in your web browser. On the left-hand menu, select **Macro Editor**.

While in the editor window, install the macro by either drag and dropping the **DWS_Node.js** on the middle pane or select **Import from file**.

*(Important) Do **NOT** enable the DWS_Node macro.*



***Required:** Repeat step 8 for ALL Node codecs.*

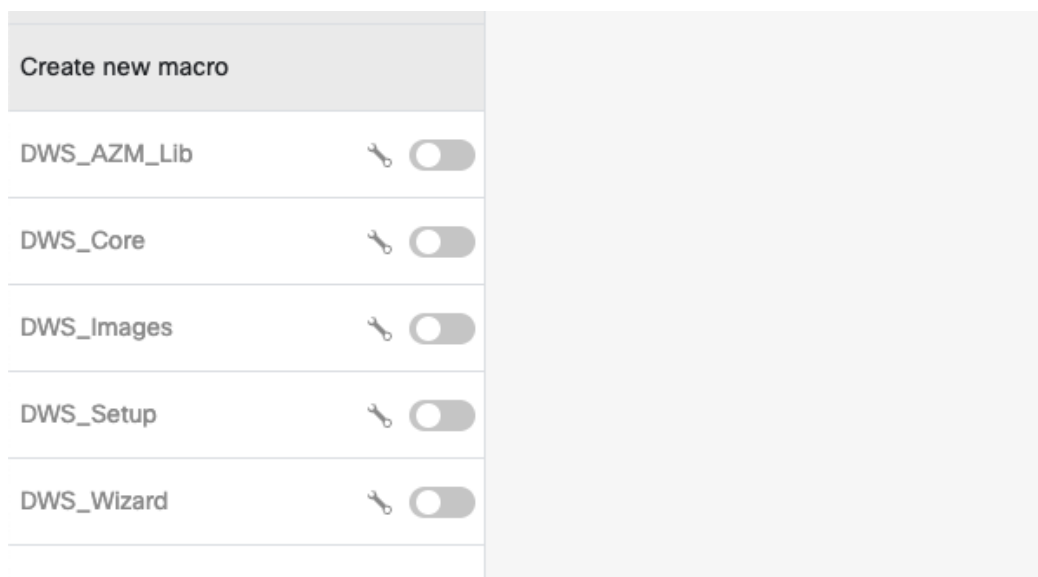
Step 9: Deploy the Wizard to the Primary Room

Log into your **Primary Room** Codec using its IP Address in your web browser. On the left-hand menu, select **Macro Editor**.

While in the editor window, install the macro by either drag and dropping the **DWS_Wizard.js** on the middle pane or select **Import from file**.

Repeat the same process for **DWS_Audio.js**, **DWS_Core.js**, **DWS_Images.js** and **DWS_Setup.js** files.

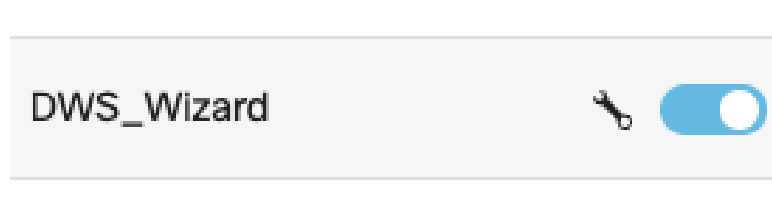
Click the Save icon next to the toggle button beside each macro file.



*Ensure all the above macros are loaded and saved before continuing.
No macro should be enabled at this step.*

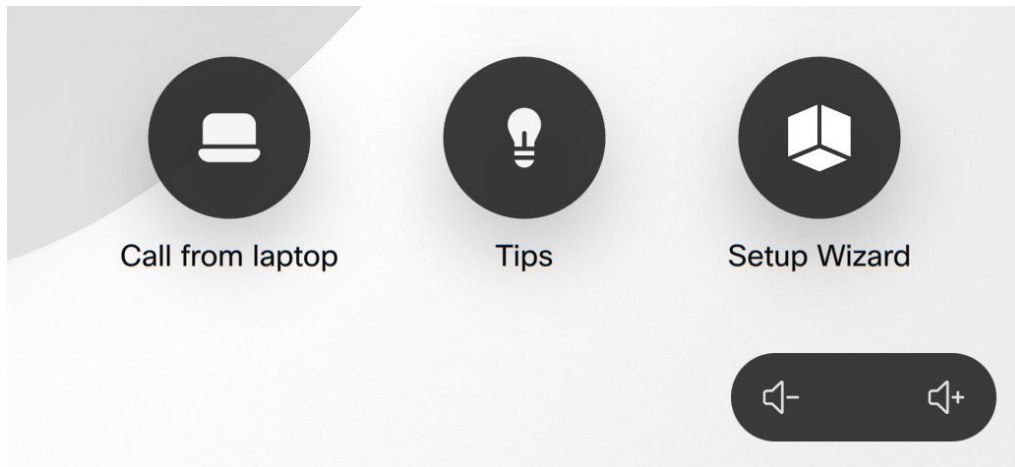
Step 10: Enable the Wizard Macro

Toggle the **DWS_Wizard** Macro to **On**.

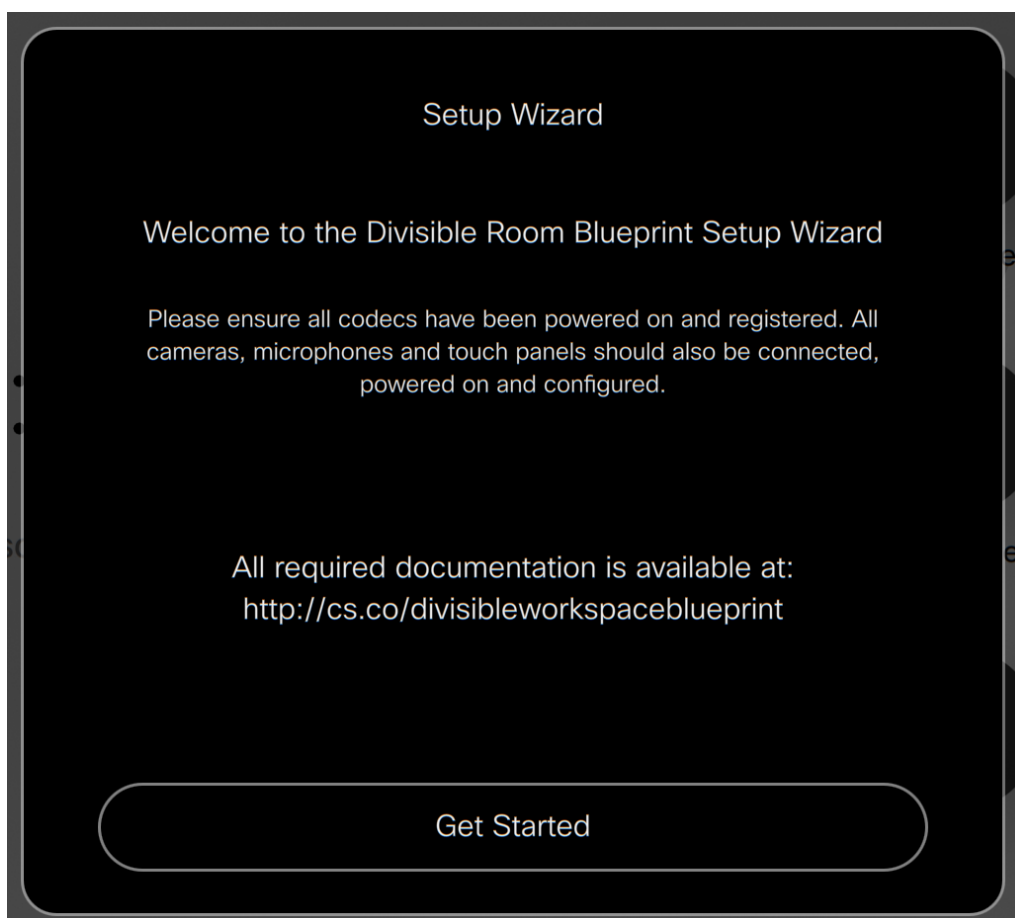


Step 11: Complete the Setup Wizard

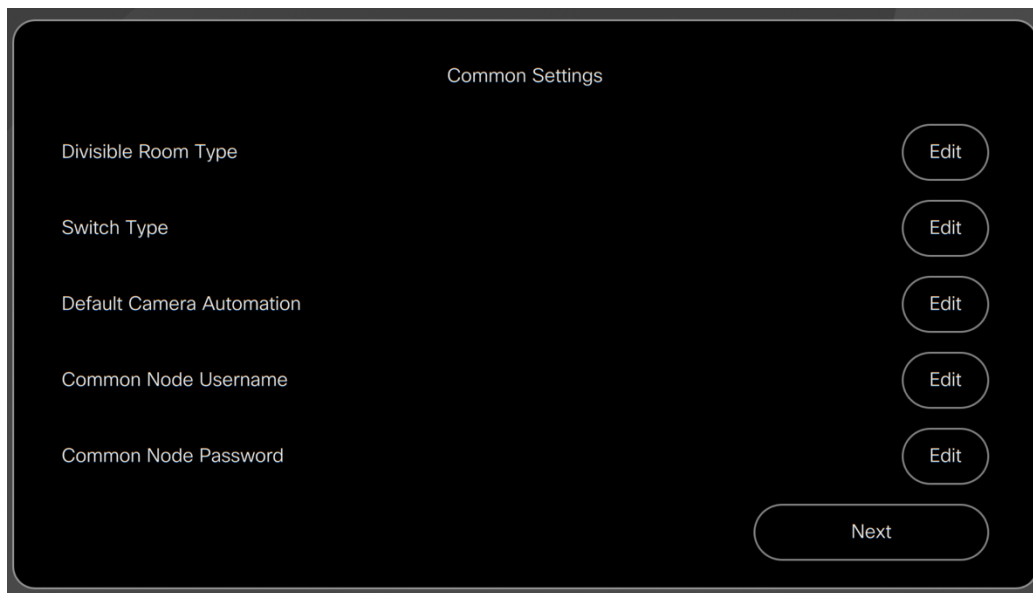
On the room Navigator, select **Setup Wizard** from the home screen.



Click “Get Started” to begin.



The first step is where your common settings will be configured. When entering the Username and Password, use the credentials you configured in Step 5.



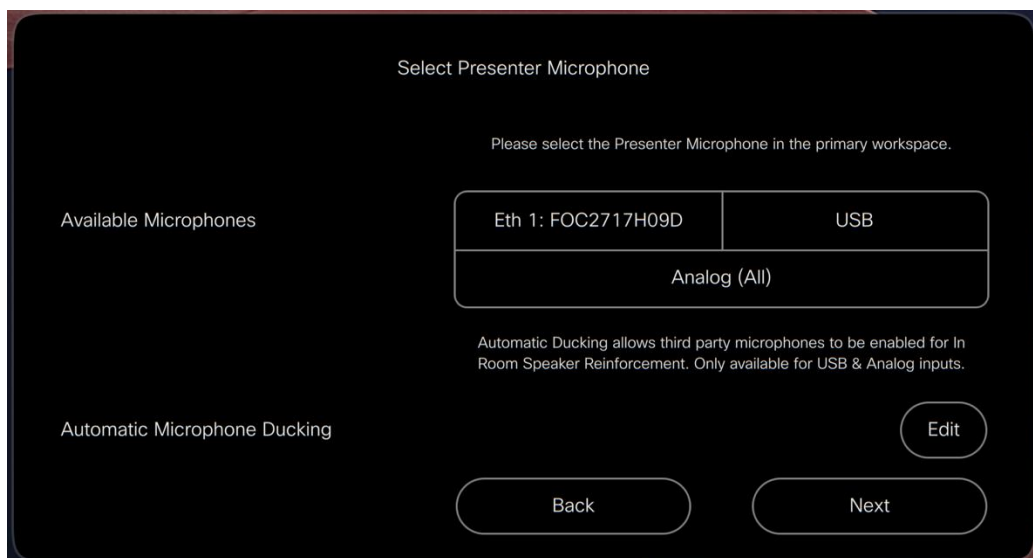
The image shows a 'Common Settings' configuration screen. It has a title bar at the top. Below the title, there are five settings listed on the left, each with an 'Edit' button on the right. The settings are: 'Divisible Room Type', 'Switch Type', 'Default Camera Automation', 'Common Node Username', and 'Common Node Password'. At the bottom right, there is a 'Next' button.

Setting	Action
Divisible Room Type	Edit
Switch Type	Edit
Default Camera Automation	Edit
Common Node Username	Edit
Common Node Password	Edit

Next

You will then be asked to select the Presenter Microphone. Cisco IP Microphones will be shown as serial numbers.

If you are using a 3rd party microphone over USB or Analog for the presenter, you can set the default for Automatic Microphone Ducking of the Cisco Pro Series microphones to support Presenter Lift scenarios.



The image shows a 'Select Presenter Microphone' configuration screen. It has a title bar at the top. Below the title, there is a instruction: 'Please select the Presenter Microphone in the primary workspace.' Below this, there is a table with 'Available Microphones' on the left and a list of microphone options on the right. The options are 'Eth 1: FOC2717H09D', 'USB', and 'Analog (All)'. Below the table, there is a text box for 'Automatic Microphone Ducking' with an 'Edit' button. At the bottom, there are 'Back' and 'Next' buttons.

Available Microphones	Options
	Eth 1: FOC2717H09D
	USB
	Analog (All)

Automatic Microphone Ducking Edit

Back Next

When you get to the Node Codec setting page, Enter the details and click **Connect**.

Node 1 Settings

Enter the details for Node 1. The Node Alias will be used on the room control panel to provide a user friendly way to ensure the correct rooms are selected.

Node 1 IP or FQDN	192.168.70.5	Edit
Node 1 Alias	Montreal	Edit
Hardware Configuration	Codec Pro with Two Screens	Edit
Presenter PTZ Type	HDMI	Edit

Back

Connect

The macro will connect to the Node Codec and gather all peripheral details. Please validate all expected peripherals have been found. If not detected, hit back, connect the peripheral and click **Connect** again.

Node 1 Details

Node 1 Codec Details	192.168.70.5
Node 1 Microphones	FOC2833JEVZ
Control Navigator	58:56:9f:7c:1d:62
Scheduler Navigator	e8:0a:b9:e3:88:4c

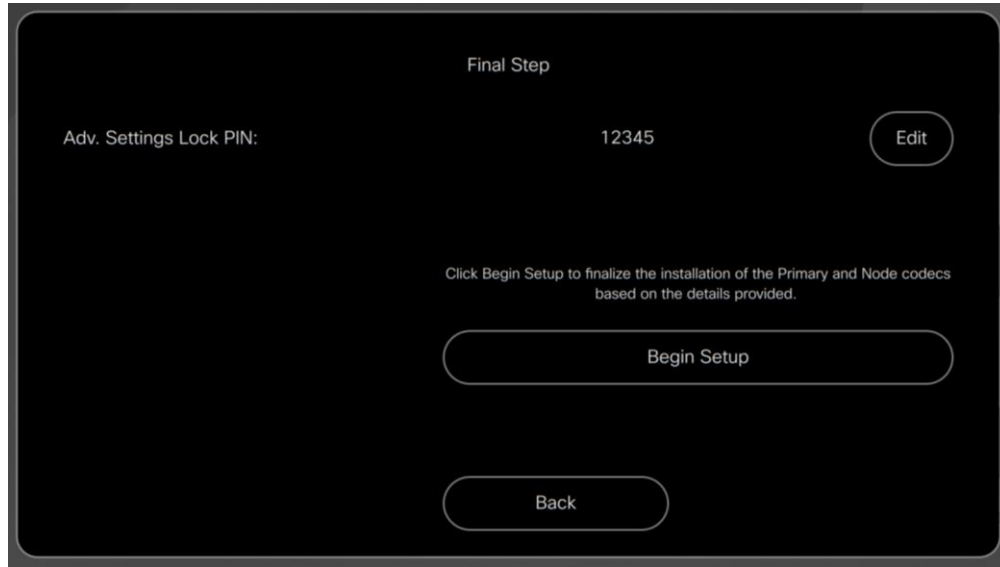
Please ensure all expected peripherals are detected before continuing.

Back

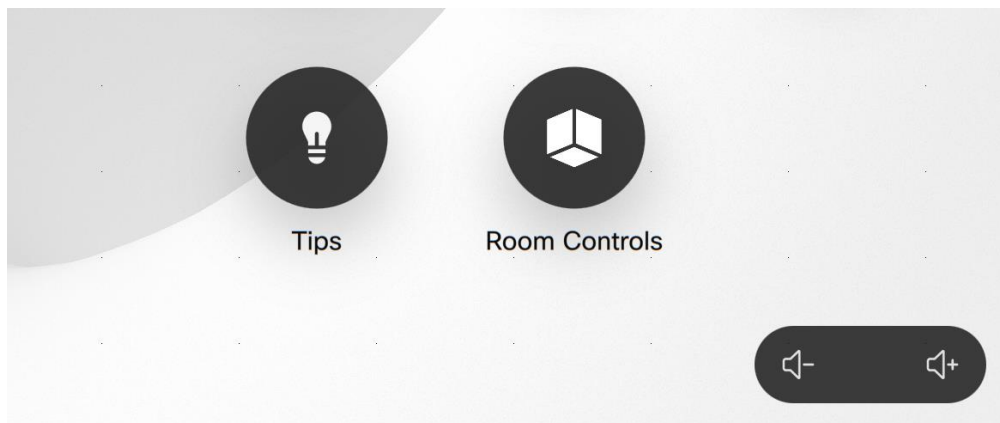
Next

If you selected Three Way Divisible you will need to repeat the steps above for Node 2.

Click **Begin Setup** to finalize your configuration.

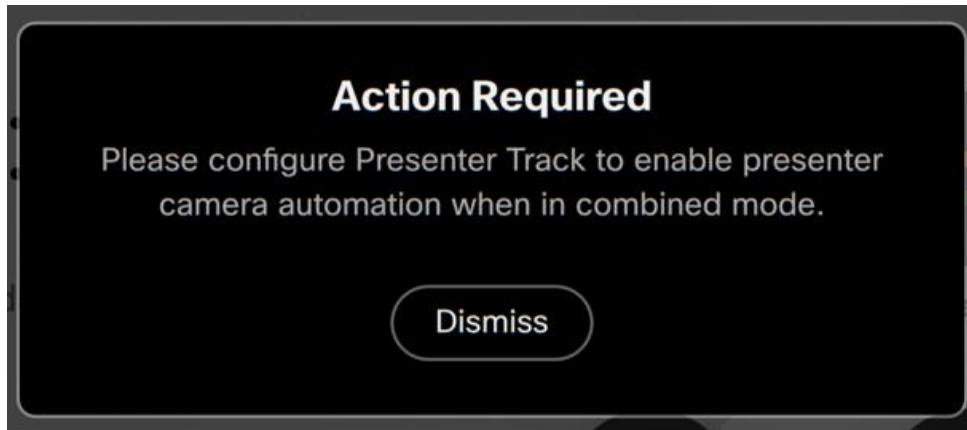


On the Room Navigator Home screen, you should now see the **Room Controls** button.



Post Wizard Configuration

You will be prompted to configure Presenter Track to enable presenter-based camera automation in combined mode operations.



Installation Complete!